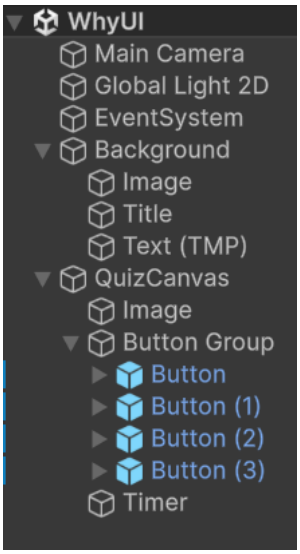
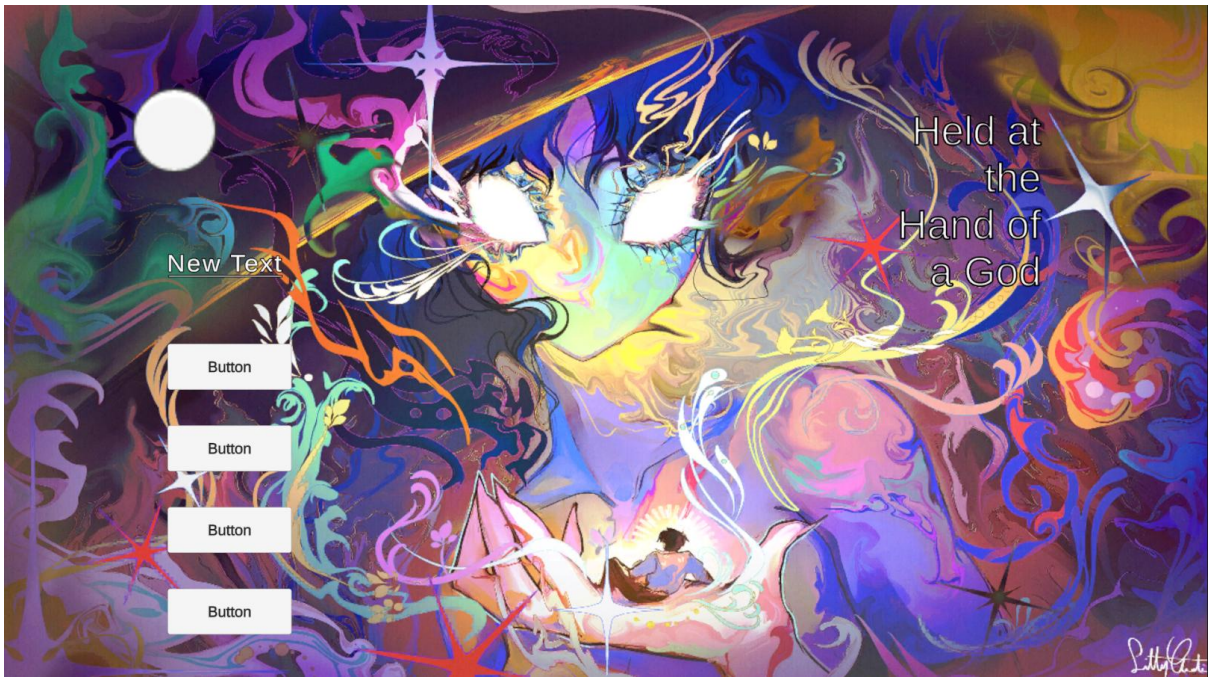
Level 1:



Img 1:All Objects present on scene.



Img 2:Aspect of UI.

Level 2:

```
[CreateAssetMenu(fileName = "Question", menuName = "New Question")]
Unity Script | 1 reference
public class QuestionSO : ScriptableObject
{
    [TextArea(2, 5)]
    [SerializeField] private string question;
    int[] nomeDoArray = new int[5];
   [SerializeField] private string[] answer = new string[4];
    [SerializeField] private int IndiceArray;


    1 reference
    public string GetQuestion()
    {
        return question;

    }
    1 reference
    public int GetAnswerCorrectIndex()
    {
        return IndiceArray;

    }
    2 references
    public string GetAnswer(int indice)
    {

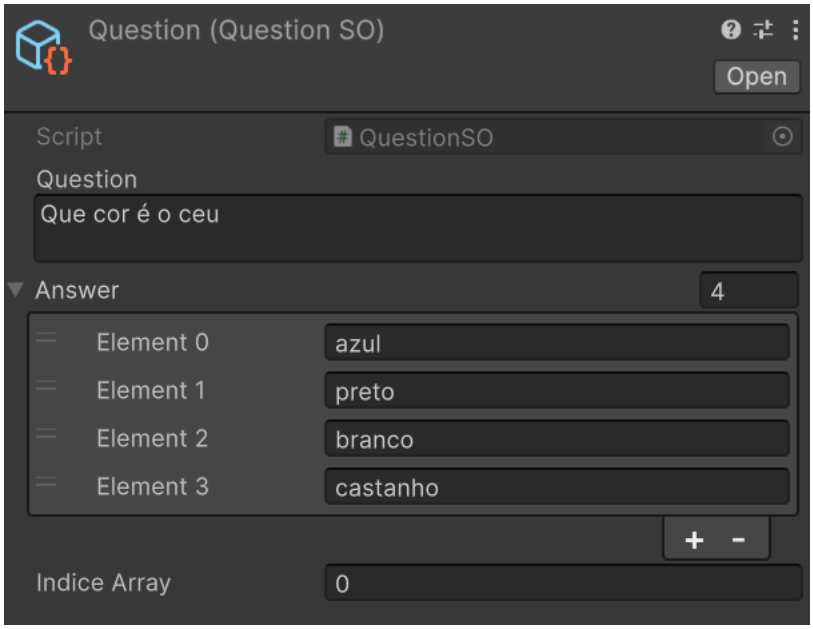
        return answer[indice];

    }


}
```

Img 3:Script Question SO that gives the question,answers and the correct answer of assigned index belonging to the question.

Level 3:



Img 4:scriptable object created with Question SO(indice Array refers to the index of the correct answer in this case its “azul”)

```
public class Questionaire : MonoBehaviour
{
    [SerializeField] private QuestionSO questionSO;
     [SerializeField] private TextMeshProUGUI textMesh;

     [SerializeField] private GameObject[] buttons = new GameObject[4];
     [SerializeField] public Timer timer;
     [SerializeField] public Image timerImage;
     private bool HasAnsweredEarly;
     private bool hasAnsweredCorrectly=false;


     // Start is called once before the first execution of Update after the MonoBehaviour is created
     Unity Message | 0 references
     void Start()
     {
         timer = FindFirstObjectByType<Timer>();
         DisplayQuestion();
         GetNextQuestion();


     }
```

```
void Update()
{
    bool teste=false;
   timerImage.fillAmount=timer.fillFraction;
    if (hasAnsweredCorrectly == true)
    {
        DisplayAnswer(0);
    }
    else
    {
        if (!timer.loadNextQuestion)
        {
            timer.loadNextQuestion = false;
            DisplayAnswer(-1);
        }
        else
        {

            timer.loadNextQuestion = true;
            GetNextQuestion();









        }
    }

    if (HasAnsweredEarly && timer.isAnsweringQuestion)
    {
        if (teste == true)
        {
            HasAnsweredEarly = false;
            DisplayAnswer(-1);
            teste = false;
        }
        else
        {
            teste = true;
            GetNextQuestion();
        }
    }

}
```

```
public void OnAnswerSelected(int index)
{
    HasAnsweredEarly = true;
    DisplayAnswer(index);
    SetButtonState(false);
    timer.CancelTimer();
}
2 references
void DisplayQuestion()
{
    textMesh.text = questionSO.GetQuestion();
    for (int i = 0; i < buttons.Length; i++)
    {
        GameObject btn_pos = buttons[i];
        TextMeshProUGUI textbtn = btn_pos.GetComponentInChildren<TextMeshProUGUI>();
        textbtn.text = questionSO.GetAnswer(i);

    }
}
3 references
void SetButtonState(bool state)
{

    for (int i = 0; i != buttons.Length; i++)
    {
        buttons[i].SetActive(state);
    }
}

3 references
void GetNextQuestion()
{
    SetButtonState(true);
    hasAnsweredCorrectly = false;
    DisplayQuestion();

}
```

```
void DisplayAnswer(int index)
{
    int GoodAnswer = questionSO.GetAnswerCorrectIndex();
    string correctAnswer = questionSO.GetAnswer(GoodAnswer);

    if (index == GoodAnswer)
    {
        textMesh.text = "Correto!!";
        hasAnsweredCorrectly=true;



    }
    else
    {

        textMesh.text = "Desculpa, mas a resposta correta é :" + correctAnswer;
        SetButtonState(false) ;


    }

}
```

Img 5:Script Questionaire controls whether the question is displayed, The answer is displayed , Makes the buttons from the button group disabled when its answered or time runs out and calls functions from Question SO to verify if correct answer given.

Level 5:

```
public class Timer : MonoBehaviour
{
    private float timerValue;
    [SerializeField] private float timeToCompleteQuestion;
    [SerializeField] private float timeToShowCorrectAnswer;
    public bool isAnsweringQuestion = false;
    public float fillFraction;

    public bool loadNextQuestion;
    // Start is called once before the first execution of Update after the MonoBehaviour is created
    Unity Message | 0 references
    void Start()
    {

    }

    // Update is called once per frame
    Unity Message | 0 references
    void Update()
    {
        UpdateTimer();
    }
```

```
void UpdateTimer()
{
    float ns = timerValue - Time.deltaTime;
    if (isAnsweringQuestion == true)
    {
        if (ns <= 0)
        {
            isAnsweringQuestion = false;
            loadNextQuestion = false;
            timerValue = timeToShowCorrectAnswer;
        }
        else
        {
            timerValue = ns;
        }
        fillFraction = timerValue / timeToCompleteQuestion;

    }
    else
    {
        if (ns <= 0)
        {
            isAnsweringQuestion = true;
            loadNextQuestion = true;
            timerValue = timeToCompleteQuestion;

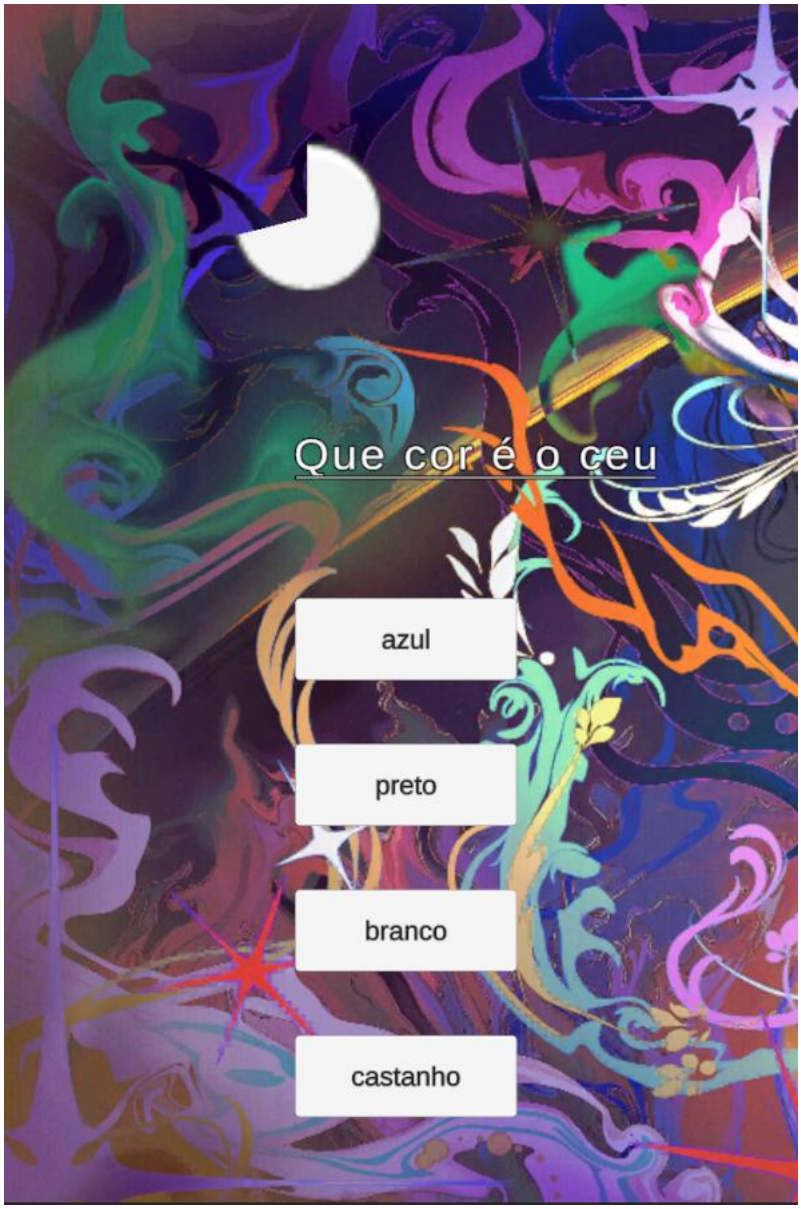
        }
        else
        {
            timerValue = ns;
        }
        fillFraction = timerValue / timeToShowCorrectAnswer;

    }

    Debug.Log(timerValue);
}


public void CancelTimer()
{
    timerValue = 0;
}
```

Img 6:Script Timer controls the timer that we can see on the ui(the white ball).



Img 7:game when starting shows you the first question(in this case since we only have one it only shows this one) and the timer starts counting down.



Img 8:this page only appears if:the timers has run out or if the user has responded incorrectly .



Img 8:This is the last state that the page can get that it just appears if the user has answered correctly.